AI4MH Crisis Signal Escalation Logic Design

GSoC 2026 Contributor Selection Task – ISSR, The University of Alabama, Mentor: David M. White,MPH,MPA

Author: *Hari Prasath* | ***Email:*** k.hariprasath.ix.c@gmail.com

Project: AI-Powered Behavioral Analysis for Mental Health Crisis Monitoring (AI4MH)

Organization: *Human ai -- ISSR Project*

Executive Summary: Engineering for Public Health Governance

During the 2025 GSoC phase, the AI4MH project successfully built a text classification pipeline , transitioning from raw Reddit/Twitter/Kaggle data into a unified dataset.Thus this resulting in BERT-based-classifier (`train_bert_from_unified.py`) and VADER especially for (`sentiment_risk_classifier.py`) sentiment risk analysis provide excellent predictions, heavily validated through SCIKIT learn metrics like roc-auc_score and precision_recall_fscore.

My contributions to AI4MH so far ive raised a couple of issues and pr's which identifies the missing file "final_report" and navigation links Issue and my PR resolve this completely and ive also identified that the run_piepline.py is empty which breaks the orchestration string also explained that in my PR

My logic_layer exactly fits above our previously built pipeline, which crossed the prototype phase, this makes it easy to upgrade our prototype.

Crisis Signal Design (Core Component)

1. **Crisis scoring framework:**

1.1 Sentiment intensity:

The phase 1 prototype only returns VADER's compound keyword like positive/negative/neutral, this fails in real world context so instead of using the normalized compound , the logic layer extracts VADER's continuous negative value and applies a scalar multiplier based on the pipeline's existing risk classification. This outputs a quantified sentiment intensity.

Code:

```
def calculate_sentiment_intensity(text, keyword_risk_level):
```

```
        vader_scores = sia.polarity_scores(text) #Extract
    negative weight
        neg_intensity = vader_scores['neg'] #Range: 0.0-1.0
         if keyword_risk_level == "High-Risk":
           multiplier = 1.5
         elif keyword_risk_level == "Moderate Concern":
            multiplier = 1.2
         else:
            multiplier = 0.5
         intensity_score = min(neg_intensity * multiplier, 1.0)
         return intensity_score
```

1.2 Volume Spike Detection:

Rather than counting the posts we actually check whether the 72-hour data is spiked from the average of 30 days previous data using normalized Z-Score, where Z-Score finds the difference in how far the observed data(72 hr cycle) is from the mean of the dataset which means the 30 day cycle.

$Z = (X-\mu) / \sigma$ OR

VSS = (current_volume − baseline_avg_volume) / baseline_std_dev

Then VSS is normalized from 0-1

VSS	Meaning
<0.2	normal
0.2–0.5	moderate increase
>0.5	abnormal spike

1.3 Geographic Clustering:

Local behavioral health crises (natural disasters) often cross county lines. We use the LAT and LON coordinate data already generated by the pipeline's “`Extract_locations.py`” script. If a target county is spiking,and If a neighboring county is *also* experiencing a volume of high-risk posts, we apply 20% boost to the Crisis Score. This tells the Health Office that they are dealing with a geographical event rather than an isolated county risk.

CODE:

```
regional_spillover = any(neighbor.high_risk_volume >
neighbor.historical_avg for neighbor in county.bordering_counties)
# Apply a 20% priority boost if the crisis is crossing county lines

regional_multiplier = 1.20 if regional_spillover else 1.00
```

1.4 Minimum Sample Size:

We establish a dynamic threshold based on the county's population, with an absolute minimum of N>25 flagged posts. Below this, the system should go to “monitoring” state. Above this should give a “high-risk” signal.

CODE:

```
dynamic_threshold = county_population * 0.0001   #THIS 0.01% IS FOR URBAN, BY PREVENTING HUGE SPIKES
 required_minimum = max(25, dynamic_threshold) #THIS MAX OF 25, FOR RURAL
    if flagged_post_count > required_minimum:
                return "EVALUATE_FOR_HIGH_RISK"
    else:
               return "MONITORING_STATE"
```

1.5 Stabilization Size

We apply exponential moving average (ema), which splits the 72 hr data into 12 hr modules and applies .ewa() method , this isolates the heavy 2-hr spikes of negative emotions (ex; late-night spiking) while providing a sustained signals

CODE:

```
# Resample 72h data into 12h buckets modules
smoothed_volume =
df.set_index('timestamp').resample('12h').size().ewm(span=3,
adjust=False).mean().iloc[-1]
```

1.6 Confidence & Uncertainty

Instead of relying on complex variance formulas, we calculate system confidence based on the model's actual certainty. A post with a BERTprob_suicide score of 0.51 is technically flagged as positive, but it represents high uncertainty. A score of 0.95 represents high certainty with high risk.

Our system confidence is simply the ratio of *highly certain* posts within the spike, minus a flat penalty if the sample size is barely above our minimum threshold.

2. Governance & Risk Controls

2.1 Bot amplification

The pipeline currently uses “clean_text_series” to strip URLs and drop basic duplicates. To prevent coordinated bot accounts from artificially increasing the volume spikes, we apply a strict **Account-Level Cap**. If a single user ID generates more than 10% of a county's high-risk volume within 72 hours, their excess posts are ignored

2.2 Media-driven spikes

When a celebrity tragedy occurs, people mostly quote news articles, causing a massive spike in suicide-related keywords. Because the prototype already imports spaCy in “Extract_locations.py” the logic layer simply extracts PERSON(celebrity) and ORG (news channels/Social media) entities from the spike. If a single named entity appears in >30% of the flagged posts, the system tags the alert with MEDIA_CONTAMINATION and pauses automated escalation.

2.3 Rural Underrepresentation

Setting a strict volume threshold accidentally excludes low-population rural counties, leaving them entirely unmonitored. As defined in our signal design, the threshold dynamically scales with population `max(25, pop * 0.0001)`, if urban county then the population will be high so max() returns pop*0.1% if rural max() returns 25

For counties that never cross the 25-post minimum, we use **Macro-County Clustering**, which means the system groups 2 or 3 adjacent rural counties together so their combined population reaches a statistical baseline, ensuring rural citizens are still represented on the state dashboard.

2.4 Human-in-the-Loop (HITL)

Health crises are human crises; machines should not make the final call.

- System Monitor (Score < 2.0): Normal, The system silently takes care of this..
- Internal Review (Score 2.0 - 2.5): The county turns yellow on the ISSR dashboard, alerting data analysts to keep an eye on it.
- The HITL Gateway (Score > 2.5): Automated escalation halts. Utilizing the pipeline's existing logic for exporting error analyses (e.g., generating “_FP_samples.csv”), the system fetches a random sample of 15 high-risk posts. An ISSR analyst must visually read the text to confirm the context (e.g., ensuring teenagers aren't just sharing sad song lyrics).

State Action: Only after the analyst clicks "Verify", the report is securely transmitted to the State Behavioral Health Office.

2.4 Audit Logging Structure

Every decision that crosses the "Internal Review" threshold generates an immutable JSON log. This builds reporting format already established in scripts like “`eval_large_csv.py`” This ensures full accountability

JSON


```
{
  "timestamp_utc": "2026-06-15 14:30:00",
  "county_id": "ALABAMA-073",
```

```
  "system_action": "ESCALATED_TO_HITL_QUEUE",
  "trigger_metrics": {
    "sample_size": 42,
    "crisis_score": 2.8,
    "system_confidence": 0.82
  },
  "noise_filters": {
    "bot_volume_suppressed": 4,
    "media_contamination_detected": false
  },
  "hitl_audit": {
    "reviewed_by_user": "issr_analyst_01",
    "decision": "APPROVED_FOR_STATE",
    "review_timestamp": "2026-06-15 14:45:00"
  }
}
```


Governance Reflection

- **Primary Risk of Premature Deployment:** The most significant risk is **automated false-escalation**, which can lead to "fake-alarm" for state health officials and the misallocation of emergency mental health resources based on "media noise" or song lyrics rather than actual crises.
- **Single Most Important Safeguard:** The **Human-in-the-Loop (HITL) Gateway** remains the critical safeguard; by preventing automated escalation at high-risk thresholds (Score > 2.5), it ensures that an ISSR analyst must verify context before any state-level action is triggered.

AI Use Disclosure

- **Original Logic:** All engineering frameworks, including **Z-Score volume spikes** and **EMA smoothing** , are original contributions designed for the **AI4MH** pipeline.
- **Tool Usage:** Google Gemini was used strictly as a technical editor for sentence refinement and document formatting to meet page constraints. Everything was verified manually